

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.(BTL 3)*

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I Nikhil Kumar Reddy Kadiveti (192425031) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Smart Healthcare Diagnosis	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	Student Academic Advising	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.
3	Intelligent Loan Approval	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.
		(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun).

		(vi) Provide the final inference with justification.
4	Vehicle Fault Diagnosis	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.
5	Cybersecurity Threat Detection	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.

Vehicle Fault Diagnosis

An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service.

Vehicle Fault Diagnosis - Expert System Architecture

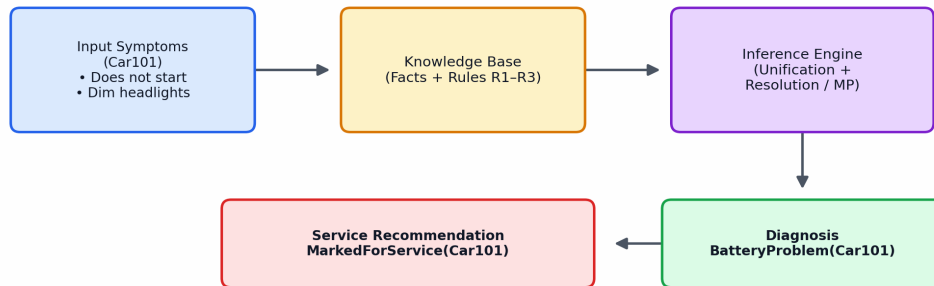


Figure 1: Expert-system architecture used for Car101's diagnosis – from input symptoms through the knowledge base and inference engine to the final service recommendation.

1. Problem Analysis and Knowledge Identification

Task (i): The scenario is analysed to extract the entities, facts, rules, predicates, variables and constants relevant to Car101's diagnosis, summarised below.

Category	Item	Description
Entity	Car101	The vehicle under diagnosis
Fact	Car101 does not start	Observed symptom 1
Fact	Car101 headlights are dim	Observed symptom 2
Rule 1	Does not start $\wedge$ dim headlights $\rightarrow$ battery problem	Diagnostic rule linking symptoms to a fault
Rule 2	Battery problem $\rightarrow$ requires battery inspection	Action rule
Rule 3	Requires battery inspection $\rightarrow$ marked for service	Service-scheduling rule
Predicate	DoesNotStart(x), DimHeadlights(x), BatteryProblem(x), RequiresInspection(x), MarkedForService(x)	Unary predicates over vehicles
Variable	x	Ranges over all vehicles
Constant	Car101	Specific vehicle instance

2. Logical Representation and Mapping (Propositional Logic)

Task (ii): Since the scenario concerns one specific vehicle (Car101), each statement can first be mapped onto propositional symbols as follows.

Symbol	Meaning
p	Car101 does not start
q	Car101's headlights are dim
r	Car101 has a battery problem
s	Car101 requires battery inspection
t	Car101 is marked for service

Propositional Formulas

Facts: p , q

Rule 1: $(p \wedge q) \rightarrow r$

Rule 2: $r \rightarrow s$

Rule 3: $s \rightarrow t$

These formulas capture the diagnostic chain purely at the level of Car101, but they cannot generalise to other vehicles — this motivates moving to First-Order Logic (FOL) in the next section.

3. Predicates, Variables, Constants & Quantifiers (FOL Representation)

Task (iii): To generalise the rules to any vehicle, we define unary predicates over a variable x (any vehicle) and the constant Car101, and quantify the rules universally so they apply to the entire fleet.

Predicates

DoesNotStart(x) – vehicle x does not start

DimHeadlights(x) – vehicle x's headlights are dim

BatteryProblem(x) – vehicle x has a battery problem

RequiresInspection(x) – vehicle x requires battery inspection

MarkedForService(x) – vehicle x is marked for service

Facts (Grounded, using constant Car101)

DoesNotStart(Car101)

DimHeadlights(Car101)

Universally Quantified Rules

R1: $\forall x \ (\text{DoesNotStart}(x) \wedge \text{DimHeadlights}(x)) \rightarrow \text{BatteryProblem}(x)$

R2: $\forall x \ \text{BatteryProblem}(x) \rightarrow \text{RequiresInspection}(x)$

R3: $\forall x \ \text{RequiresInspection}(x) \rightarrow \text{MarkedForService}(x)$

The universal quantifier ($\forall$) is used because each rule must hold for every vehicle in the service center's fleet, not just Car101; no existential quantifier is required since the rules are unconditional generalisations.

4. Logical Connectives and Formula Construction

Task (iii cont.): Each rule is built using standard logical connectives, applied consistently between the propositional and FOL forms:

- $\wedge$ (AND) – combines DoesNotStart(x) and DimHeadlights(x) as joint evidence in R1.
- $\rightarrow$ (IMPLICATION) – encodes each diagnostic/action rule (R1, R2, R3): antecedent symptoms imply a consequent conclusion.
- $\neg$ (NOT) – used when forming clausal form for resolution (Section 7), e.g. $\neg$ DoesNotStart(x).
- $\vee$ (OR) – used when each implication is rewritten as a disjunctive clause, e.g. $R1 \equiv \neg$ DoesNotStart(x) $\vee \neg$ DimHeadlights(x) $\vee$ BatteryProblem(x).

No equivalence ($\leftrightarrow$) is needed here since all relationships in the scenario are one-directional diagnostic implications, not biconditional definitions.

5. Inference and Logical Reasoning

Task (v/vi, forward chaining view): Starting from the two facts about Car101, Modus Ponens is applied repeatedly through R1, R2 and R3 to derive the final conclusion.



Figure 2: Step-by-step inference chain – facts about Car101 are propagated through Rules 1–3 by Modus Ponens to reach the final conclusion.

Reasoning Steps

- Step 1: DoesNotStart(Car101) and DimHeadlights(Car101) are given as facts.
- Step 2: Applying R1 by Modus Ponens → BatteryProblem(Car101) is derived.
- Step 3: Applying R2 by Modus Ponens → RequiresBatteryInspection(Car101) is derived.
- Step 4: Applying R3 by Modus Ponens → MarkedForService(Car101) is derived.

Final Conclusion: Car101 has a battery problem, requires a battery inspection, and is marked for service.

6. Unification and Substitution

Task (iv): Unification is required to match the general (variable-based) predicates in the rules with the specific (grounded) facts about Car101. Two representative unifications are shown below.

Unification 1

DoesNotStart(x) $\equiv$ unify with $\equiv$ DoesNotStart(Car101)

Most General Unifier (MGU): $\theta = \{ x / \text{Car101} \}$

Unification 2

DimHeadlights(x) $\equiv$ unify with $\equiv$ DimHeadlights(Car101)

Most General Unifier (MGU): $\theta = \{ x / \text{Car101} \}$

Since both predicates in R1 share the same variable x, the same substitution $\theta = \{x/\text{Car101}\}$ is applied consistently across the whole rule, producing the grounded rule instance:

DoesNotStart(Car101) $\wedge$ DimHeadlights(Car101) $\rightarrow$ BatteryProblem(Car101)

This grounded rule is what allows the facts about Car101 to be directly resolved against Rule 1 in the next step.

7. Resolution and Proof Construction

Task (v): Resolution refutation is used to formally prove BatteryProblem(Car101). All statements are first converted into clausal (CNF) form, the goal is negated, and resolution is applied until the empty clause ($\emptyset$) is derived, confirming the goal by contradiction.

Clause Set

C1: DoesNotStart(Car101)	[fact]
C2: DimHeadlights(Car101)	[fact]
C3: $\neg \text{DoesNotStart}(x) \vee \neg \text{DimHeadlights}(x) \vee \text{BatteryProblem}(x)$	[Rule 1, clausal form]
$\neg G$: $\neg \text{BatteryProblem}(\text{Car101})$	[negated goal]

Resolution Steps

- Resolve C1 with C3 using $\theta = \{x/\text{Car101}\}$: cancel DoesNotStart(Car101) $\rightarrow$ C4: $\neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$
- Resolve C4 with C2: cancel DimHeadlights(Car101) $\rightarrow$ C5: BatteryProblem(Car101)
- Resolve C5 with $\neg G$: cancel BatteryProblem(Car101) $\rightarrow \emptyset$ (empty clause)

The empty clause is a contradiction, so by refutation the original goal BatteryProblem(Car101) is proved true. The same resolution pattern, applied to R2 and R3 in turn, proves RequiresBatteryInspection(Car101)

and finally `MarkedForService(Car101)`.

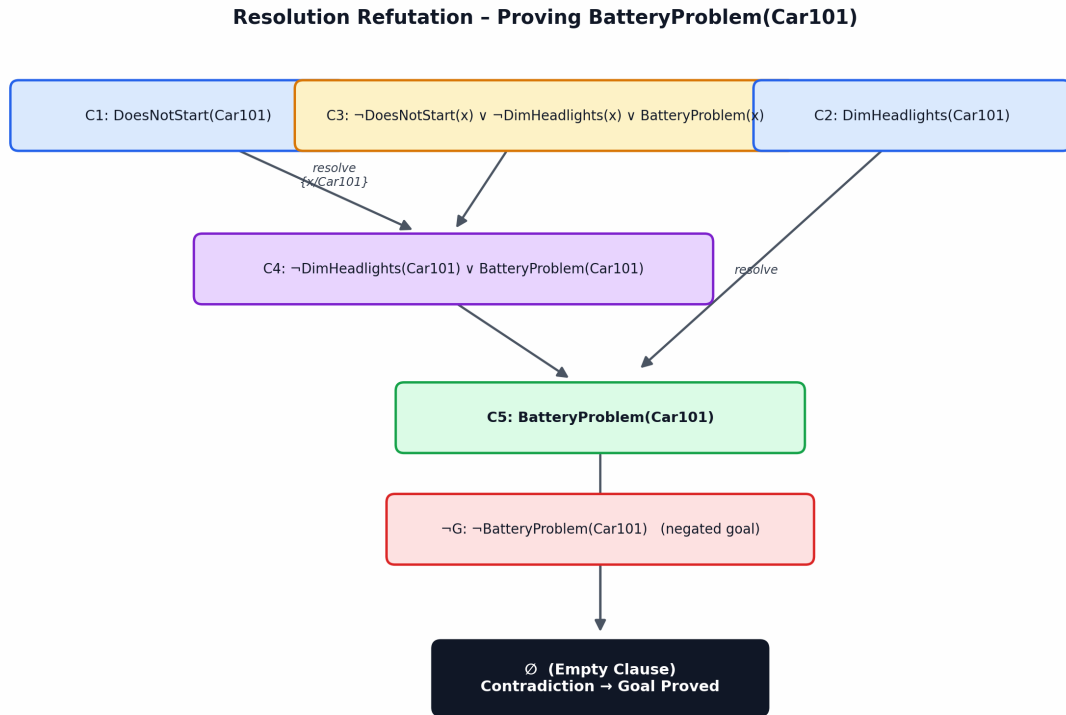


Figure 3: Resolution refutation tree – clauses C1–C3 and the negated goal are resolved down to the empty clause ($\emptyset$), proving `Car101` has a battery problem.

8. Final Inference, Mapping Diagram & Conclusion

Task (vi): Combining the forward-chaining derivation (Section 5) and the resolution proof (Section 7), the complete reasoning chain for `Car101` is:

`DoesNotStart(Car101) ∧ DimHeadlights(Car101)`

⇒ (R1) `BatteryProblem(Car101)`

⇒ (R2) `RequiresBatteryInspection(Car101)`

⇒ (R3) `MarkedForService(Car101)`

Final Conclusion: `Car101` does have a battery problem, and it should be marked for service. The full knowledge-mapping diagram (Figure 2) and resolution tree (Figure 3) together provide the clear, well-organised visual proof of this inference, satisfying both the reasoning and presentation requirements of the exercise.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.
		construct valid			

		logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.